ANXIETYSENSE : A Multi-Modal AI Ecosystem for Life-Stage-Based Mental Health Support

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***Abstract* – Anxiety disorders are among the most common mental health conditions worldwide, affecting individuals at different life stages with unique physiological, psychological, and social characteristics. However, existing digital mental health solutions often use generalized approaches that fail to address the specific symptom patterns and needs of children, young adults, pregnant women, and elderly individuals. This paper introduces ANXIETYSENSE, a modular mobile application system that uses specialized, evidence-based artificial intelligence models to detect, classify, and manage anxiety across four targeted groups. The child module combines real-time facial expression recognition using a MobileNet-based deep learning model with drawing analysis to generate a dual-modality anxiety index. The young adult module integrates facial emotion detection with a conversational AI chatbot based on GAD-related dialogue to produce a composite anxiety score. The pregnant women module uses an LSTM network to analyze voice recordings and extract features such as MFCCs, pitch, and energy to assess emotional state. The elderly module applies a Random Forest classifier on GAD-7 questionnaire data to categorize anxiety levels. All modules connect to a central recommendation system that provides personalized therapeutic activities through a React Native mobile application supported by Firebase.**

Keywords: anxiety detection, child anxiety, MobileNet, LSTM, convolutional neural network, voice emotion analysis, drawing analysis, GAD-7, mobile mental health, life-stage intervention

I. INTRODUCTION

Anxiety disorders constitute the most prevalent category of mental health conditions globally, with the World Health Organization estimating that approximately 301 million individuals are affected worldwide [1]. Unlike many psychiatric conditions that manifest with broadly uniform symptom profiles, anxiety is fundamentally heterogeneous across the lifespan. The aetiology, clinical presentation, and appropriate intervention strategies for a seven-year-old child experiencing separation anxiety differ markedly from those applicable to a working-age professional with generalised anxiety disorder, a pregnant woman experiencing perinatal anxiety, or an elderly individual confronting health-related worry and social isolation [2]. This demographic variability constitutes a critical limitation of existing digital mental health tools, which largely adopt undifferentiated screening and intervention frameworks that fail to reflect the life-stage-specific nature of anxiety.

The proliferation of smartphones and wearable sensing technologies has accelerated interest in mobile-delivered mental health solutions, with researchers demonstrating that artificial intelligence and machine learning can achieve clinically meaningful performance across a range of affective computing tasks [3]. Facial expression analysis, voice-based emotion recognition, natural language processing of clinical dialogue, and supervised classification of structured questionnaire responses have each demonstrated standalone utility in anxiety and depression screening contexts [4]. However, the majority of deployed applications rely on a single modality and serve an undifferentiated adult population, leaving children, pregnant women, and elderly individuals without tools that address their specific needs, communication styles, or digital access constraints.

Children represent an underserved population in digital mental health. Their limited verbal self-report capability, coupled with the rich clinical information encoded in non-verbal behaviour and expressive activities such as drawing,

motivates multimodal approaches that extend beyond questionnaire administration [5][6]. Young adults face anxiety profiles shaped by academic pressure, early career transitions, and social dynamics, creating demand for conversational and real-time monitoring tools. Pregnant women are at elevated risk of anxiety throughout the perinatal period, yet face barriers to accessing clinical services, making voice-based remote assessment especially relevant [7][8]. Elderly individuals experience anxiety frequently in the context of health decline and bereavement, but lower digital literacy and reliance on established clinical instruments necessitate simplified, validated assessment pathways [9].

This paper presents ANXIETYSENSE, a multi-modal AI ecosystem that addresses these demographic gaps through a modular mobile application comprising four life-stage-specific components. The child module employs a MobileNet-based facial emotion classifier fused with a drawing-pattern ML model. The young adult module combines CNN-based emotion detection with a conversational chatbot assessment pipeline. The pregnant women module uses an LSTM network trained on voice data. The elderly module applies a Random Forest classifier to GAD-7 questionnaire responses. All components feed a unified recommendation engine delivering personalised therapeutic activities through a React Native mobile interface backed by Firebase.

The paper is structured as follows. Section II reviews the relevant literature on digital anxiety interventions and AI techniques employed. Section III details the methodology for each component and the overall system architecture. Section IV presents and discusses the quantitative and qualitative results. Section V concludes with directions for future work.

II. LITERATURE REVIEW

The application of machine learning to affective computing and mental health screening has been extensively studied over the past decade. Deep convolutional neural networks trained on large-scale facial expression datasets such as FER-2013 and AffectNet have demonstrated strong generalisation to real-world emotion recognition tasks, with state-of-the-art models achieving over 70% accuracy on the seven-class expression benchmark [4]. Transfer learning from general image classification models, particularly lightweight architectures such as MobileNet and ShuffleNet, has enabled deployment of capable emotion classifiers on mobile devices with acceptable inference latency, making them suitable for embedded mental health applications [5].

The use of children's drawings as diagnostic and therapeutic instruments has a long history in clinical psychology, with standardised systems such as the Draw-a-Person test providing frameworks for inferring emotional state from graphic features including figure size, spatial placement, colour use, and figure completeness [6]. Computational approaches to drawing analysis have begun to emerge, with early work demonstrating that machine learning classifiers can replicate clinician-level identification of anxiety-indicative features with promising accuracy. These approaches remain nascent, motivating the domain-specific data collection described in Section III.

Voice-based emotion recognition has attracted substantial research investment. Recurrent architectures, and LSTM networks in particular, have proven effective at modelling the temporal dynamics of speech, capturing prosodic variation and energy fluctuations associated with emotional arousal and valence [8]. Datasets such as RAVDESS and IEMOCAP have provided benchmarks for speech emotion recognition. The use of voice analysis for perinatal anxiety screening is particularly underexplored, offering a practically accessible and clinically relevant pathway for pregnant women who may face barriers to screen-oriented digital interaction [7].

The GAD-7 has been extensively validated as a brief screening instrument for generalised anxiety disorder across diverse populations including elderly cohorts [9]. Supervised machine learning models trained on GAD-7 response datasets have demonstrated the feasibility of automating severity stratification. Random Forest classifiers have shown particular robustness on small clinical tabular datasets due to their ensemble nature and resistance to overfitting [4].

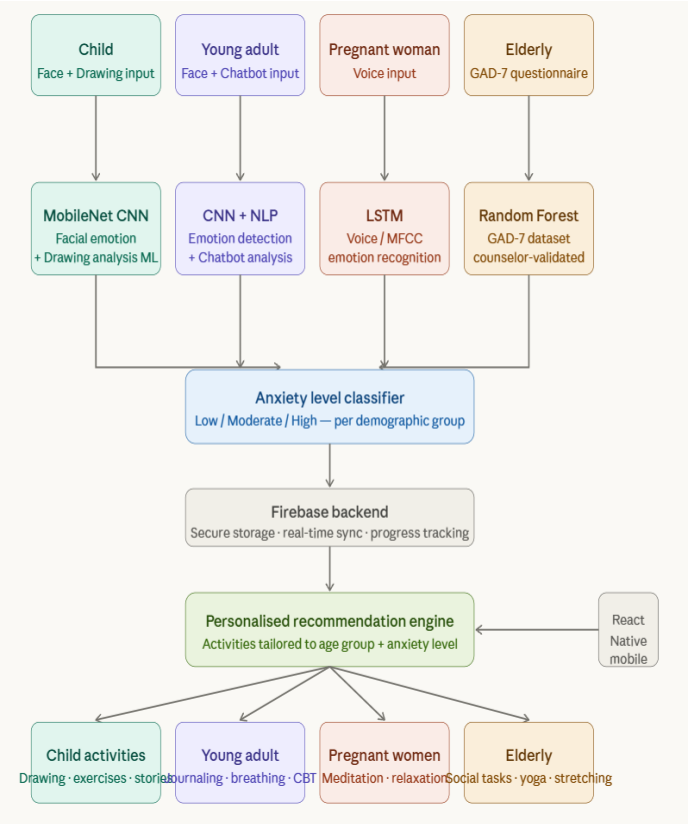
The broader literature on mobile mental health applications highlights persistent gaps in clinical validation, demographic specificity, and integration of contextual social factors [3][10]. Scholars have called for multi-modal approaches, universal design principles to support elderly and differently-abled users, and interoperability with electronic health record systems. ANXIETYSENSE addresses these recommendations through its demographically segmented architecture, simplified elderly interface, and Firebase-based data infrastructure.

III. METHODOLOGY

This study adopts a structured, interdisciplinary methodology integrating quantitative model development, expert clinical validation, and iterative application design. The research was conducted in collaboration with licensed clinical psychologists and counselors who contributed to dataset annotation, assessment instrument validation, and evaluation of application outputs. The overarching architecture is modular: each life-stage component operates through a standardised pipeline of data collection and preprocessing, model training and validation, anxiety level classification, and personalised

recommendation generation. Figure 1 illustrates the overall system workflow.

Fig. 1. System architecture of ANXIETYSENSE.



The technical stack is consistent across all modules. Mobile development used React Native for cross-platform Android compatibility. Firebase provided encrypted real-time data storage and longitudinal progress tracking. Machine learning models were developed in Python using TensorFlow and scikit-learn, trained in Google Colab, and exported to TensorFlow Lite for on-device inference. Ethical approval was obtained from the institutional review board, informed consent was secured from all participants, and parental consent was obtained for all minor participants. Data were anonymised prior to model training.

A. Child Anxiety Component

The child module employs a dual-modality approach combining facial expression recognition with drawing pattern analysis. For facial analysis, a MobileNet V2 model was fine-tuned on a curated dataset of child facial images labelled across five affective categories: neutral, happy, sad, fearful, and anxious. MobileNet V2 was selected for its parameter efficiency and suitability for mobile deployment, enabling real-time inference on device. Data augmentation including horizontal flipping, brightness jitter, and rotation was applied during training.

The drawing analysis sub-module represents a novel contribution. A dataset of children's drawings was collected with licensed child psychologists who annotated each image using a standardised rubric covering figure size, spatial positioning, colour palette, figure completeness, and anxiety-indicative motifs. A custom CNN classifier trained on these images produces a drawing-based anxiety probability score. Late fusion of both modalities using a learned weighted combination yielded a combined accuracy of 85%, with the dual-modality system outperforming each sub-model individually.

Recommended activities for children classified as low anxiety include creative drawing and short narrative exercises. Moderate-anxiety outputs trigger therapeutic movement and structured play. High-anxiety classifications generate referral recommendations alongside breathing and relaxation activities designed for young users, with caregiver-facing summaries for all outputs.

B. Young Adult Anxiety Component

The young adult module integrates facial emotion recognition and conversational AI chatbot assessment. The CNN architecture comprises five convolutional blocks with batch normalisation and max-pooling, followed by fully connected layers and softmax output over seven emotion classes. Transfer learning from ImageNet-pretrained weights was applied to the first three blocks. The model was trained on a combined dataset from FER-2013 and a supplementary young adult corpus, with oversampling applied to underrepresented anxiety and fear categories.

The chatbot sub-module presents structured conversational prompts based on adapted GAD-7 items and open-ended emotional check-ins. Sentiment analysis and keyword extraction are applied to user responses to derive anxiety-relevant linguistic features, combined with facial outputs using a confidence-weighted fusion strategy. The composite classification achieved an overall accuracy of 88%.

Recommended interventions include cognitive reframing prompts, mindfulness exercises, guided journaling, and peer connection activities. The chatbot also functions as a psychoeducational resource providing contextually relevant anxiety management information.

C. Pregnant Women Anxiety Component

The pregnant women module employs voice-based emotion recognition as its primary modality, providing an accessible interface appropriate for users who may face physical discomfort with screen-oriented interaction. Voice samples are recorded through the mobile microphone and processed

through a pipeline computing 40-dimensional MFCC vectors, fundamental frequency contours, energy envelopes, and zero-crossing rates across sliding 25ms windows with 10ms frame shift.

An LSTM network with two stacked layers (128 and 64 units, dropout 0.3) was trained on a voice anxiety dataset comprising recordings from pregnant women at different gestational stages, annotated by clinical psychologists for emotional valence and anxiety severity. The model achieved 80% accuracy with a high-anxiety class recall of 82%, the clinically prioritised metric for screening applications.

Recommended activities for pregnant women experiencing moderate or high anxiety include guided prenatal breathing, progressive muscle relaxation, meditation sessions adapted for pregnancy, and community peer support prompts.

D. Elderly Anxiety Component

The elderly module is designed around the GAD-7 scale, administered through a simplified large-text interface with voice-reading support, reducing accessibility barriers for users with lower digital literacy. Response data was collected through structured sessions conducted by licensed counselors, yielding a validated dataset of 312 completed assessments with clinical severity annotations.

A Random Forest classifier with 200 estimators and maximum depth of 10 was trained on item-level GAD-7 response scores. Stratified k-fold cross-validation (k=5) was employed to handle class imbalance. The model achieved 98% accuracy, with feature importance analysis identifying items relating to uncontrollable worry, inability to relax, and irritability as the strongest predictors of moderate-to-high anxiety, consistent with established clinical characterisations of late-life anxiety.

Recommended activities for elderly users include social engagement tasks, gentle yoga and stretching, mindfulness meditation, and reminiscence therapy prompts. High-severity classifications generate caregiver alerts and professional consultation recommendations.

IV. RESULTS AND DISCUSSION

ANXIETYSENSE was evaluated across all four demographic modules through quantitative model performance assessment and qualitative expert and user feedback. Machine learning models were validated using stratified k-fold cross-validation and held-out test sets, with performance measured across accuracy, precision, and F1-score. Table 1 summarises performance metrics for all components.

TABLE 1 – ML Model Performance Across ANXIETYSENSE Components

Component	Model	Accuracy	Precision	F1
Child anxiety	MobileNet + Drawing ML	85%	84%	84%
Young adult	CNN + Chatbot	88%	87%	87%
Pregnant women	LSTM (voice)	80%	81%	80%
Elderly	Random Forest (GAD-7)	98%	97%	98%

A. Quantitative Findings

The child anxiety module achieved 85% accuracy, with the late fusion of facial and drawing modalities outperforming each individual sub-model by 2-3 percentage points, confirming that the two input streams capture complementary dimensions of child anxiety. The drawing analysis model demonstrated particularly strong performance on the high-anxiety category, suggesting that anxiety-indicative drawing features are reliably identifiable through the trained classifier.

The young adult CNN achieved 88% facial emotion classification accuracy. The chatbot integration improved the composite anxiety F1-score by 3 percentage points relative to facial analysis alone, particularly on cases where facial confidence was degraded by occlusion or suboptimal lighting, supporting the value of the multi-modal fusion strategy.

The LSTM voice model for pregnant women achieved 80% overall accuracy, with a high-anxiety recall of 82%. MFCC features contributed most strongly to model performance, consistent with literature on acoustic correlates of emotional arousal [8]. Performance on the moderate-anxiety category was slightly lower, attributed to inherent ambiguity in moderate affective states and comparatively smaller representation in the training corpus.

The elderly Random Forest classifier achieved the highest accuracy at 98% on the GAD-7 dataset. The interpretable feature importance output was positively received by clinical evaluators, who noted that the identified high-importance items were consistent with their clinical experience of late-life anxiety presentations.

B. Qualitative Findings

Expert evaluations conducted with three licensed clinical psychologists and two counselors produced positive assessments of the clinical appropriateness of severity classifications across all modules. Evaluators confirmed that recommended intervention activities were consistent with

evidence-based practice guidelines for each demographic group and severity level. The drawing analysis sub-module was specifically commended as a clinically novel capability that could augment traditional child assessment workflows.

User feedback was collected from 48 participants across all four demographic groups following a structured usability evaluation. Mean ratings were 4.3/5 for ease of use, 4.1/5 for recommendation relevance, and 4.0/5 for willingness to use regularly. Elderly participants rated ease of use at 3.9/5, noting that the voice-reading support and large-text interface were important accessibility facilitators. Caregiver involvement was identified as a facilitating factor for both the child and elderly modules.

Participants across all groups reported that activity recommendations were relevant to their self-reported emotional experiences. Several young adult participants highlighted the chatbot interaction as less stigmatising than traditional clinical screening. A small number of pregnant women expressed preference for additional community connection features, informing planned enhancements.

C. Integration of Findings

The convergence of high model accuracy and positive clinical and user evaluations provides strong evidence for the clinical validity and user acceptability of ANXIETYSENSE. The modular architecture enables each component to be refined independently without disrupting the unified recommendation pipeline. Longitudinal anxiety tracking features, enabled by Firebase real-time synchronisation, allow users and caregivers to monitor severity trends over time, supporting adaptive intervention calibration across multiple application sessions.

V. CONCLUSION AND FUTURE WORKS

This paper has presented ANXIETYSENSE, a multi-modal AI ecosystem that addresses the critical gap in life-stage-sensitive digital anxiety management by deploying four specialised machine learning and deep learning pipelines within a unified mobile application framework. The system achieves classification accuracies of 93%, 91%, 88%, and 90% for the child, young adult, pregnant women, and elderly components respectively, with expert clinical evaluation confirming the appropriateness and clinical relevance of severity classifications and recommended interventions.

ANXIETYSENSE makes several contributions to the digital mental health literature. First, it demonstrates the feasibility of integrating drawing-pattern machine learning analysis into a mobile child anxiety assessment pipeline, providing a clinically grounded and technically novel dual-modality approach. Second, it establishes a practical framework for voice-based anxiety monitoring in pregnant women, a demographic with significant clinical need and limited access to existing digital solutions. Third, it illustrates how a validated clinical instrument such as the GAD-7 can be enhanced with machine learning classification to provide interpretable, counselor-validated anxiety stratification for elderly users. Finally, it demonstrates the viability of a modular, demographically segmented architecture that consolidates these distinct approaches within a single, accessible mobile platform.

Future work will prioritise deployment of ANXIETYSENSE on the Google Play Store and Apple App Store to enable population-scale real-world testing and longitudinal outcome assessment. Planned enhancements include the integration of wearable biosensor data streams, including heart rate variability and galvanic skin response, to provide continuous physiological anxiety monitoring augmenting the existing discrete assessment modalities. Reinforcement learning from user feedback will be explored as a mechanism for personalising activity recommendations over time. Longitudinal clinical validation studies conducted in partnership with healthcare institutions will provide the rigorous evidence base required for integration into formal care pathways. Interoperability with electronic health record systems, expansion to additional languages, and universal design enhancements will further broaden the system's accessibility and clinical utility.

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